

05 — LTL Properties

LTL Operators

Op	Name	Meaning
G	Globally	Holds in every future state
F	Eventually	Holds in some future state
X	Next	Holds in the next state
U	Until	p holds until q becomes true
W	Weak until	Like U but q may never come
R	Release	Dual of U: q until $p \wedge q$
->	Implies	Material implication
&&	And	Conjunction
	Or	Disjunction
!	Not	Negation

Syntax

```
formula response {  
  over System;  
  body = ltl G(request -> F(grant));  
}
```

The `ltl` keyword triggers automatic translation to mu-calculus before evaluation.

Property Patterns

Safety — nothing bad ever happens:

```
| body = ltl G safe;
```

Liveness — every stimulus gets a response:

```
| body = ltl G (request -> F grant);
```

Response (GR1) — repeated stimulus, repeated response:

```
| body = ltl G (trigger -> F ack);
```

Persistence — eventually always stable:

```
| body = ltl F G stable;
```

Recurrence — event happens infinitely often:

```
| body = ltl G F heartbeat;
```

Precedence — q does not occur before p :

```
| body = ltl (!q) W p;
```

LTL to Mu-Calculus Translation

LTL	Mu-Calculus
$G(\text{safe})$	$\nu X. \text{safe} \wedge []X$
$F(\text{target})$	$\mu X. \text{target} \vee \langle \rangle X$
$G F(p)$	$\nu X. (\mu Y. p \vee \langle \rangle Y) \wedge []X$
$p U q$	$\mu X. q \vee (p \wedge \langle \rangle X)$
$X(p)$	$\langle \rangle p$

Key: ν = greatest fixpoint (safety/invariance),
 μ = least fixpoint (liveness/reachability),
 $[]$ = box (all successors), $\langle \rangle$ = diamond (some successor).

Controllable Attractor

Controllable reachability (game-theoretic):

```
| mu X. (p || [ (ctrl = controllable) ] X)
```

$[\text{ctrl} = \text{controllable}] X$ means:

- ALL uncontrollable transitions $\rightarrow X$
- At least ONE controllable $\rightarrow X$

When to Use LTL vs Mu-Calculus

- **Use LTL** for standard temporal patterns (safety, liveness, response) — simpler and less error-prone.
- **Use mu-calculus** when you need:
 - Label-specific modalities: $[a]$, $\langle b \rangle$.
 - Deeply nested or alternating fixpoints.
 - Properties that mix box/diamond on different labels.
- LTL formulas are translated automatically; no performance penalty.

Further Reading

Pnueli (1977) “The Temporal Logic of Programs” — foundational LTL paper. Dwyer, Avrunin & Corbett (1999) — property specification patterns. Manna & Pnueli (1992) *The Temporal Logic of Reactive and Concurrent Systems*.